

Fast Local Swap Using Processing-in-Memory A Viability Study with UPMEM

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The Memory Wall Problem

Why do we need fast swap?

Problem & Context

Results

Conclusion

Modern Applications Exhaust RAM

- Scientific computing, ML, databases, virtualization
- **Memory pressure** forces OS to swap pages to storage
- Traditional swap = **major performance bottleneck**

Performance Impact

- **HDD:** 5-10 milliseconds
- **SSD:** 60-200 microseconds

Can we do better without expensive hardware?

Existing Solutions & Their Limitations

The design space is crowded but imperfect

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Traditional Block Storage (SSD)

Pros:

- TB-scale capacity
- Well-established, reliable

Cons:

- **Slow:** 60-200 μs (SATA/NVMe I/O)
- Complete page fault: **160 μs**
- Tail latency: up to **23ms** under load

Remote Memory (InfiniSwap)

Pros:

- Fast: 30-40 μs
- Unlimited capacity

Cons:

- Complex deployment
- Network dependency

Compression (zram)

Pros:

- Fast: 20-50 μs
- No disk I/O

Cons:

- **Capacity paradox:** Consumes 33-50% RAM
- CPU overhead

Gap: Fast + Local + True Capacity Extension

Our Approach: UPMEM PIM

Combining speed with simplicity

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What is UPMEM?

- Commercial PIM
- 2,556 DPUs (160GB capacity)
- Each DPU: 64MB MRAM storage
- **Outside main RAM** = true extension

Key Insight

Use UPMEM MRAM as **fast local swap backend**

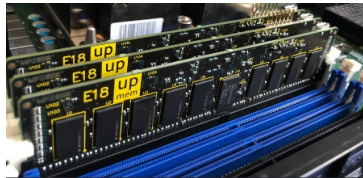


Figure: UPMEM DPUs integrated in DDR4

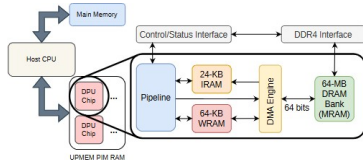


Figure: UPMEM Architecture

Design Space: Remote memory-class performance without network infra.

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Our Approach: UPMEM as Local Swap

Simple concept, powerful results - PaaS

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Core Idea

Use UPMEM's MRAM capacity as a **fast local swap tier**

- Sits between RAM and SSD in memory hierarchy
- No network infrastructure required
- True capacity extension (not RAM redistribution like zram)

What We Evaluated

- **Latency:** How fast can we swap 4KB pages?
- **Scalability:** Does using more DPUs help? (1-64 DPUs tested)
- **Efficiency:** What's the sweet spot configuration?

Experimental Approach

Simulation-based viability study using UPMEM SDK simulator
(98.4% correlation with real hardware - validated by prior work)

Key Result: Fast & Predictable Latency

Competitive with remote memory, no network needed

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Main Achievement

30-50 μ s

Complete swap latency
(end-to-end, 4KB page)

Performance vs SSD

- Traditional SSD: 60-200 μ s
- UPMEM: 30-50 μ s
- **2-5 $\times$ faster!**

Why This Matters

**For memory-intensive
edge apps:**

- Real-time video analytics
- Edge ML inference
- IoT data processing
- Local databases

Multi-DPU Scaling & Sweet Spot

Parallel mode unlocks significant speedup

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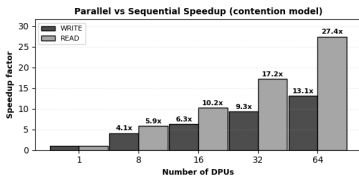


Figure: Speedup: Parallel vs Serial

Scaling Analysis

- **8 DPUs:** 6× speedup, 75% efficiency
- **16 DPUs:** 10× speedup, 65% efficiency
- **64 DPUs:** 27× speedup, 42% efficiency

Efficiency Formula

Efficiency = Speedup / Number of DPUs

Sweet Spot

8-16 DPUs optimal

- Best efficiency (65-75%)
- Latency: 35-42 μ s

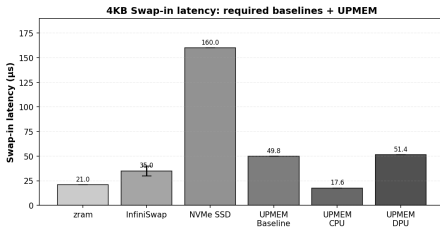
Comparison with State-of-the-Art

UPMEM fills a critical gap

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Solution	Latency	Network
UPMEM	30-50 μ s	No
InfiniSwap	30-40 μ s	Yes
zram	20-50 μ s	No*
SSD(NVMe)	60-160 μ s	No

*Consumes 33-50% RAM

Figure: Latency Comparison (4KB page)

UPMEM's Unique Position

- **vs InfiniSwap:** Same speed, no network, simpler deployment
- **vs zram:** Same speed, no RAM consumption, true capacity extension
- **vs SSD:** 2-5 $\times$ faster

Key Takeaways

- **Viability demonstrated:** UPMEM achieves 30-50 μ s swap latency
- **Sweet spot identified:** 8-16 DPUs balance performance & efficiency
- **Fills critical gap:** Local deployment with remote memory-class speed
- **Practical solution:** No InfiniBand, no RAM consumption penalty

Target Deployments

- **Standalone systems:** No datacenter infrastructure available

Next Steps

- Hardware validation with real UPMEM system
- Production kernel module and integration

Thank You for your attention..

Questions ?

